On the possibility of an analytic expression for the extremal mass
of equal-spin rotating black holes in five-dimensional Einstein–Gauss–Bonnet gravity

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Question

Is there an analytic expression for the extremal mass $M_{\rm ext}(J,\alpha)$ of the five-dimensional, asymptotically flat, equal-spin ($J_1 = J_2 = J$) Einstein–Gauss–Bonnet black hole, action $(16\pi G_5)^{-1}\int\sqrt{-g}\,(R+\alpha\mathcal{L}_{\rm GB})$, $G_5 = 1$? Scale invariance reduces the problem to one function of one variable,

$$M_{\rm ext} = \pi^{1/3} J^{2/3}\, m(s), \qquad s = \frac{\pi^{2/3}\alpha}{J^{2/3}}, \qquad \text{equivalently} \qquad \mu(y) = \frac{M_{\rm ext}}{J^{2/3}} = \pi^{1/3} m(\pi^{2/3} y), \quad y = \frac{\alpha}{J^{2/3}}, \tag{1}$$

and the slope studied in the manuscript is $\partial M_{\rm ext}/\partial\alpha|_J = \mu'(y) = \pi\, m'(s)$.

Short answer

A closed form for $m(s)$ is not available and, on the present evidence, is not feasible with the tools used in this project. What *is* established, and what this review verified independently, is an exact description of both ends of the function and of the near-horizon data at every coupling, together with a global representation accurate to 10^{-5}. No new closed form was found here; one further natural attempt (expansion in the algebraic throat variable) was made and is reported below as a negative result.

What is known exactly

Weak coupling. The project's exact exterior solver (`extremal-mass/code/pert.py`) gives, with $\zeta_n = \zeta(n)$,

$$\begin{aligned} m(s) &= \tfrac{3}{2} + s - \tfrac{236}{105}s^2 + d_3 s^3 + d_4 s^4 + d_5 s^5 + d_6 s^6 + \dots, \qquad (2)\\ d_3 &= \tfrac{60851752}{33075} - 1536\,\zeta_3 = -6.548307\dots, \qquad d_4 = -\tfrac{592541765354296}{1489863375} + \tfrac{227328}{5}\zeta_3 + \tfrac{88064}{25}\pi^4 = 66.100522\dots,\\ d_5 &= \tfrac{447853140007115845664}{6100990520625} + \tfrac{657983488}{175}\zeta_3 - \tfrac{3972366336}{625}\zeta_2^2 - \tfrac{1464139776}{25}\zeta_5\\ &= 423.920197\dots,\\ d_6 &= -4737.494827\dots \quad \text{(exact form in } \texttt{extremal-mass/series/coefficients.json}\text{)}. \end{aligned}$$

The first two terms are the published result of Ma, Li and Lü. The coefficients are exact conditional on the identification (PSLQ at 100 digits) of regularised horizon constants with multiple zeta values; the audit found that this identification is over-determined by the horizon-regularity conditions of the solver (injecting a wrong value for one constant makes the order-3 run abort with "h not regular"), and the rational horizon shifts X_5, X_6 coincide with an independent algebraic prediction. The series is therefore trustworthy. Its radius of convergence is small: the throat data have a square-root branch point at $s_* = 0.00592 \pm 0.05482\,i$, $|s_*| = 0.0551$; the six known d_n are compatible with, but do not establish, the same radius for $m(s)$ (the ratio test on six terms only bounds it by roughly 0.12).

Near-horizon data at any coupling. Writing $g_H = r_H^2$, $a = \alpha/g_H$ and $q = h_H/g_H$ for the horizon squashing, Sen's entropy function on the throat $\mathrm{AdS}_2 \ltimes S^3$ gives (rederived here from scratch, 12 assertions in `checks/check_throat2.py`)

$$16a^2q^2(4-q) - 8a(2q^2-7q+4) + q - 2 = 0, \qquad \frac{8J^2}{\pi^2 g_H^3} = \frac{q^2(4-q)\,[1+4a(4-3q)]}{8}, \tag{3}$$

$$S_{\rm ext} = \frac{\pi^2}{2}\, g_H^{3/2}\sqrt{q}\,\big[1+4a(4-q)\big],$$

$$v_1 = \frac{(1+4aq)(12aq-16a-1)}{4(6aq^2-16aq-1)}, \qquad e^2 = \frac{(q-4)(1+4aq)^2(12aq-16a-1)}{4q\,(6aq^2-16aq-1)^2},$$

with v_1 the AdS_2 radius squared and e the fibre electric field. At fixed J these equations determine $g_H(s)$, $q(s)$ and $S_{\rm ext}(s)$ as algebraic functions. They do *not* determine the mass: $M_{\rm ext}$ needs the asymptotic normalisation of the time coordinate, which the throat discards. (Equation (3.18) of Kleihaus–Kunz–Radu prints the total $J_1 + J_2$; the project's per-plane normalisation is the consistent one.)

Strong coupling. Matched asymptotics and, independently, a BTZ dictionary for the small-r core of static EGB black holes give

$$m(s) = \frac{3s}{4} + \frac{4}{3}\, s^{-1/2} + a_2\, s^{-1} + a_3\, s^{-3/2} + \dots, \qquad M_{\rm ext} = \frac{3\pi\alpha}{4} + \frac{4J}{3\sqrt{\alpha}} + a_2 \frac{J^{4/3}}{\pi^{1/3}\alpha} + \dots \tag{4}$$

The leading term is the mass of the nakedly singular static solution (the endpoint conjectured by Kleihaus, Kunz and Radu); $a_1 = 4/3$ is derived under the assumption that the rotating extremal core obeys the static BTZ dictionary and is confirmed numerically to 4×10^{-4}; $a_2 = 0.456527$ is a conditional analytic value, numerically 0.4566 ± 0.0002; $a_3 \simeq -0.263$ is purely numerical. The audit noted that the BTZ relation, taken literally beyond leading order, would give $a_2 = 0$, so the dictionary is a leading-order statement only.

Exact identities that constrain any candidate. The Smarr relation at $T = 0$ gives $M_{\rm ext} = 3\Omega_H J + \alpha\, \partial M_{\rm ext}/\partial\alpha|_J$, i.e. $3\,\Omega_H J^{1/3} = \mu - y\mu'$, so the minimum of the slope is the maximum of the scaled horizon angular velocity; the first law at $T = 0$ gives $\partial M_{\rm ext}/\partial J|_\alpha = 2\Omega_H$. Both hold on all three numerical branches checked here.

Numerical knowledge, cross-validated

Three independent determinations exist in the repository: the manuscript's zero-temperature extrapolation of non-extremal solutions (13 couplings, $s \le 1.10$), the direct $T = 0$ spectral solver of `work/kkr-extremal` ($s \le 5.2$), and the shooting / multiple-shooting branch of `extremal-mass/strong` ($2 \le s \le 103$). This review compared them with each other, with the exact series and with the ten-term global approximant (table 1, figures 1–2): m agrees to 6×10^{-6} relative across all sources, the slope to 3×10^{-4} (within the manuscript's own fit envelopes), and the minimum of the slope is $\partial M_{\rm ext}/\partial\alpha|_J \simeq 1.070$ at $y \simeq 0.19$ ($s \simeq 0.40$) on every route. The approximant

$$m(s) \simeq \frac{3s}{4} + \sum_{k=1}^{10} c_k\, t^k, \qquad t = \left(1 + \frac{s}{\kappa}\right)^{-1/2}, \quad \kappa = 0.356242695, \tag{5}$$

with

$$c_{1\dots5} = 2.233910,\ 1.281505,\ -0.713812,\ 6.055449,\ -75.189708,$$
$$c_{6\dots10} = 230.148117,\ -352.252482,\ 303.906361,\ -142.214324,\ 28.244984,$$

reproduces $m(0) = 3/2$, $m'(0) = 1$, $a_1 = 4/3$, $a_2 = 0.456527$ exactly and every numerical point to 1.1×10^{-5}. It is the best available "expression" for practical use, but it is an interpolant: its Taylor coefficients beyond s^1 and its $s^{-3/2}$ coefficient (-0.15 instead of -0.26) are not those of $m(s)$.

Why a closed form is unlikely, and what was tried

1. **Transcendental structure.** d_n contains multiple zeta values of weight up to n (ζ_3 at order 3, ζ_2^2 at order 4, ζ_5 at order 5, ζ_3^2 and ζ_2^3 at order 6). An algebraic function of s over $\mathbb{Q}$, or a hypergeometric-type function of s with rational parameters, has rational Taylor coefficients. Zeta values of growing weight arise instead when the *parameters* of such functions depend on s, for instance through Γ-functions of the throat exponent $\gamma(s)$, which is exact and irrational ($\lambda^2 + \lambda = Q(a, q)$ in the project's endpoint report). Any closed form would therefore have to be non-elementary, with s entering through exponents.

2. **Singularities.** The weak series has complex singularities close to the origin (conjecturally s_*, radius 0.055); the function has an essential change of behaviour at $s \to \infty$ (expansion in $s^{-1/2}$ after a linear term). A simple formula interpolating the two ends and respecting s_* was sought by the previous session without success (finite-Laurent Hamilton–Jacobi superpotentials, Liouvillian solutions of the squashing equation — its Galois group is $SL(2, \mathbb{C})$ — and a resonant candidate ODE for $m(s)$, all ruled out).

3. **New attempt in this review: the throat variable.** Since s_* is a critical point of the exact algebraic map $s(q)$, the fold disappears in the variable q; both ends of the branch are series in q ($q \to 2$ at Myers–Perry, $q \to 0$ at strong coupling, with $s^{-1/2} \propto q$ there). We composed the exact series with $s(q)$ and expanded

$y(q) = m - 3s/4$ in $w = 2 - q$: $y = \frac{3}{2} + 0.03125\,w - 0.05595\,w^2 + 0.10348\,w^3 - 0.26867\,w^4 + 0.65415\,w^5 - 1.80098\,w^6 + \ldots$ The coefficients grow geometrically with ratio ≈ -2.6, i.e. radius $|w| \approx 0.4$, and the exact discriminant $D(q) = 5q^4 - 34q^3 + 73q^2 - 56q + 16$ of the throat vanishes at $q = 2.2763$, $|2 - q| = 0.276$: on the branch continued to negative coupling q has an extremum there, so q is not a uniformising coordinate near Myers–Perry. Against the numerics the w-series is *worse* than the s-series (error 2.6×10^{-3} at $s = 0.044$ versus 7.5×10^{-6}), and a [3/3] Padé in w fails beyond $s \approx 0.05$. A [3/3] Padé of $m(s)$ itself extends the useful range only to $s \approx 0.4$ (error 5×10^{-3}) and is useless beyond $s \approx 0.7$. Negative result; receipts in `checks/receipts/uniformize_q.txt`.

What would move the question

(i) An exact derivation of a_2 from the next order of the core/intermediate matching, which would fix the strong-coupling side to two exact terms; (ii) one or two more exact orders d_7, d_8 (the solver scales badly: order 6 took 1 h 38 min and 200 MB) to test the conjectured radius $|s_*|$ and the large-order form $d_n \sim 2\,\mathrm{Re}[C\, s_*^{-n} n^{\beta}]$; (iii) a search for a resummed structure in which s enters through the exact throat exponent $\gamma(s)$, e.g. ratios of Γ-functions reproducing the weight pattern of the d_n; (iv) a proof, from the horizon expansion in $r^{2\gamma}$, that $m(s)$ inherits the throat fold. None of these is a small task, and none is guaranteed to yield a closed form. Until then the honest statement is: $M_{\mathrm{ext}}(J, \alpha)$ is known exactly as two asymptotic expansions, its near-horizon data are algebraic at every coupling, and the function in between is known numerically to 10^{-5} and summarised by eq. (5).

comparison (worst over the common range)	quantity	deviation
manuscript $T \to 0$ vs exact weak series ($s < 0.055$)	m	4.9×10^{-6} rel.
manuscript $T \to 0$ vs global approximant (13 rows)	m	6.4×10^{-6} rel.
manuscript $T \to 0$ vs direct $T = 0$ solver	μ	2.4×10^{-5}
manuscript Ψ_{ext} vs $\pi m'$ of the approximant	slope	2.1×10^{-3}
manuscript Ψ_{ext} vs direct $T = 0$ solver μ'	slope	3.0×10^{-4}
multiple-shooting table ($2 \le s \le 103$) vs approximant	m	6.9×10^{-6} rel.
minimum of the slope: approximant / direct solver / manuscript lowest sample	μ'	1.0700 (y=0.185) / 1.0698 (y=0.195) / 1.0715 (y=0.175)

Table 1: Cross-validation of the three independent numerical determinations (`checks/check_branch_consistency.py`).

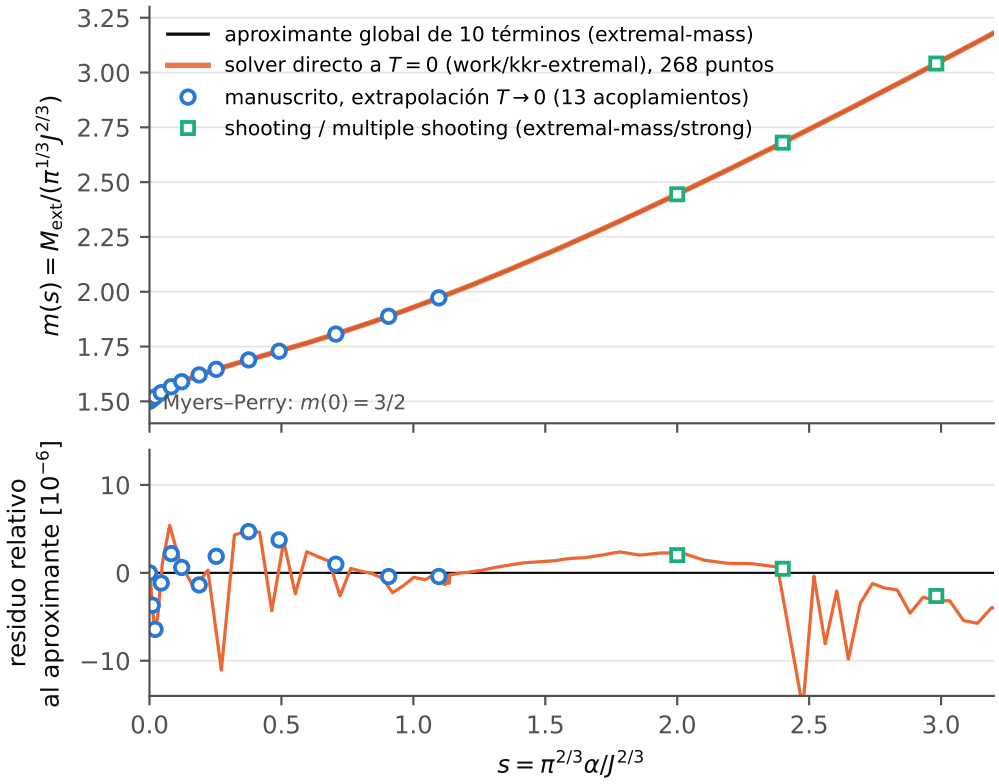


Figure 1: The three numerical routes and the global approximant; lower panel: relative residuals.

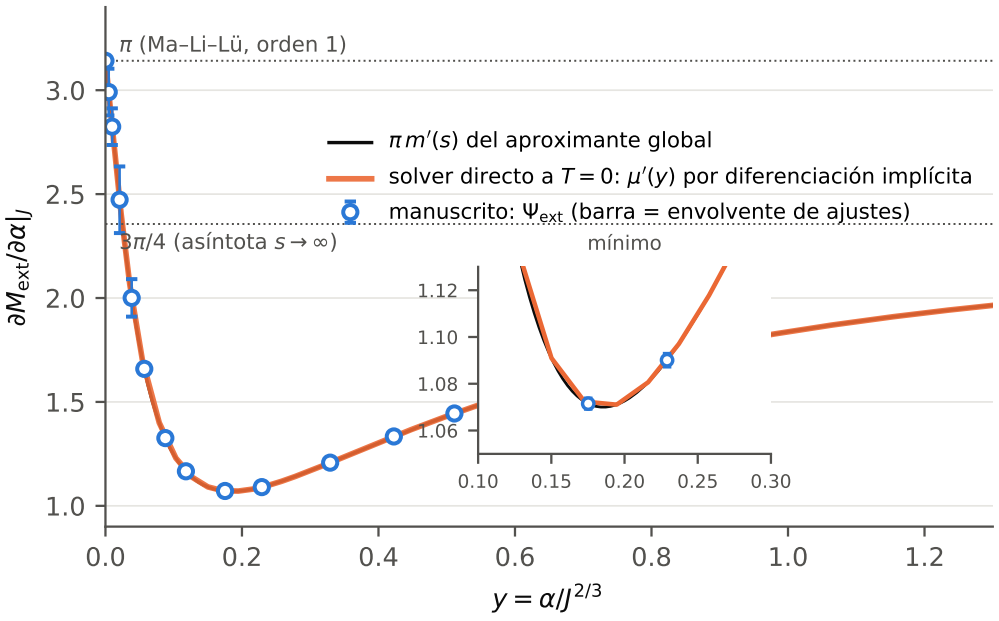


Figure 2: The slope $\partial M_{\rm ext}/\partial\alpha|_J$ by the three routes.